



Booth's Algorithm

Binary Multiplication – Booth's Algorithm:

- Multiplies two signed binary numbers in two's complement notation
- Invented by Andrew Donald Booth in 1950.



CONDITIONS:

1. If $Q_0 Q_{-1}$ are same i.e. 00 or 11 then, perform arithmetic right shift by 1 bit.
2. If $Q_0 Q_{-1} = 10$ then perform

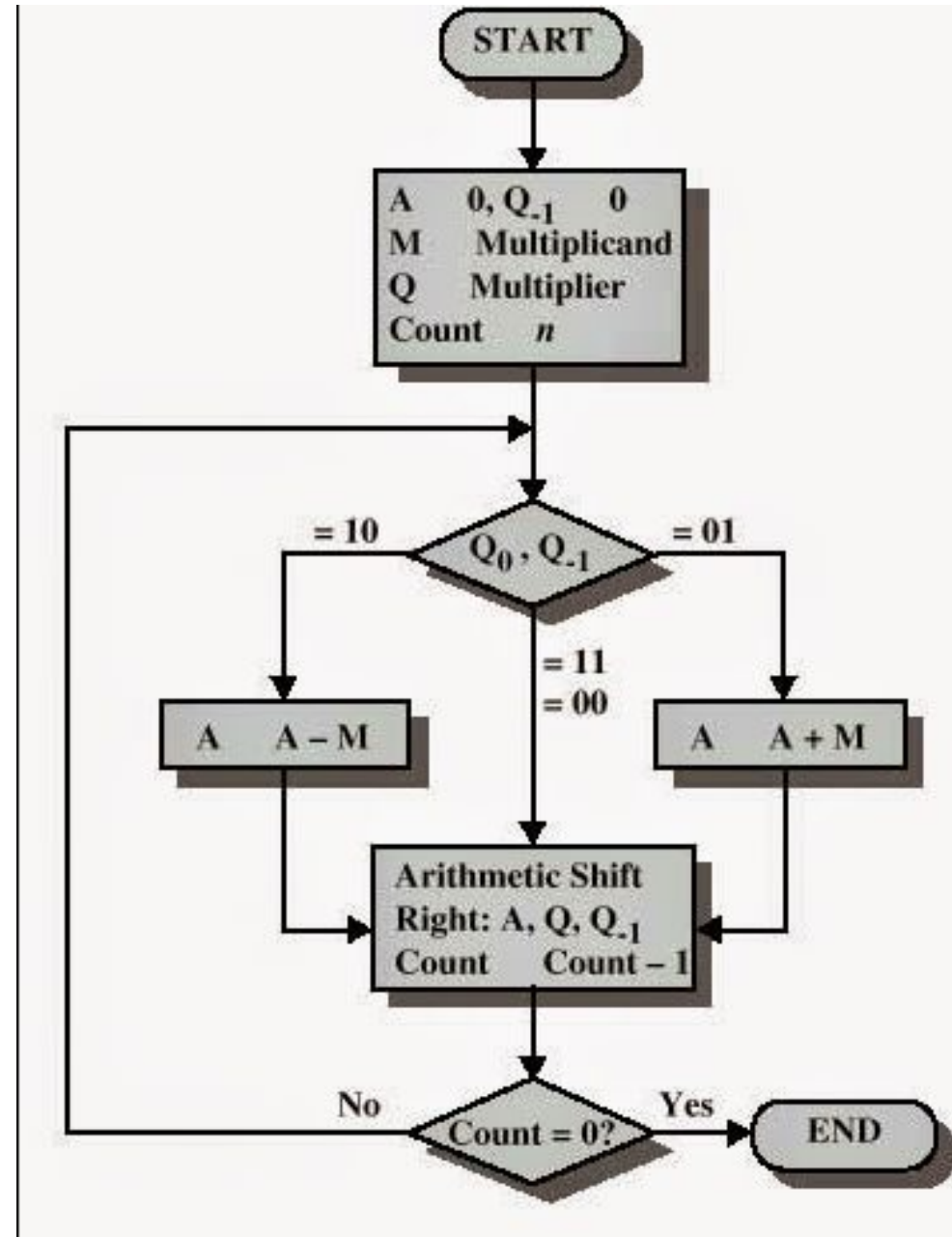
$$A \leftarrow A - M$$

And then perform arithmetic right shift.

3. If $Q_0 Q_{-1} = 01$ then perform

$$A \leftarrow A + M$$

And then perform arithmetic right shift.



Binary Addition

$$0 + 0 = 0$$

$$0 + 1 = 1$$

$$1 + 0 = 1$$

$$1 + 1 = 0 \text{ with 1 carry}$$

$$1 + 1 + 1 = 1 \text{ with 1 carry}$$

$$\begin{array}{r} 1 \\ + 1 \\ \hline 2 \end{array} \rightarrow \underline{10}$$

Binary Addition

Weight	16	8	4	2	1
		1	0	1	0
+		1	1	1	0
	1	1	0	0	0

(10)

(14)

(24)

Binary Multiplication – Booth's Algorithm:

Multiplicand(M): 0 1 0 1 1

Multiplier(Q): 0 1 1 1 0

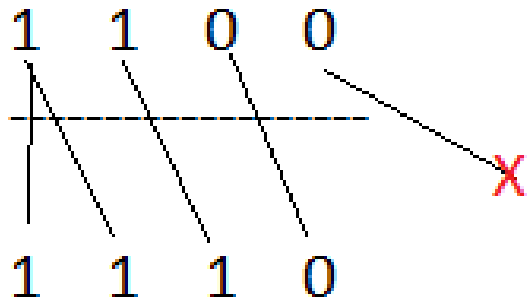
We will check Q_0 and Q_{-1} values.

initialize $Ac=0$, $Q_0=0$;

Q_{-1} is last value of Q

Count = 4 (represents how many bits)

Arithmetic Shift Right



Whenever we do ASR make count - -

Multiply 7 and 3 Using Booth's Algorithm

Multiplicand $M = 7$ Binary Value : 0 1 1 1

Multiplier $Q = 3$ Binary Value : 0 0 1 1

Count 4 Bits

$AC = AC + M$ (Binary Addition already we learnt)

$AC = AC - M$

If we want to avoid subtraction

We will convert this $AC = AC - M$ to $AC = AC + (-M)$

Now we need to find $-M$ Binary Value

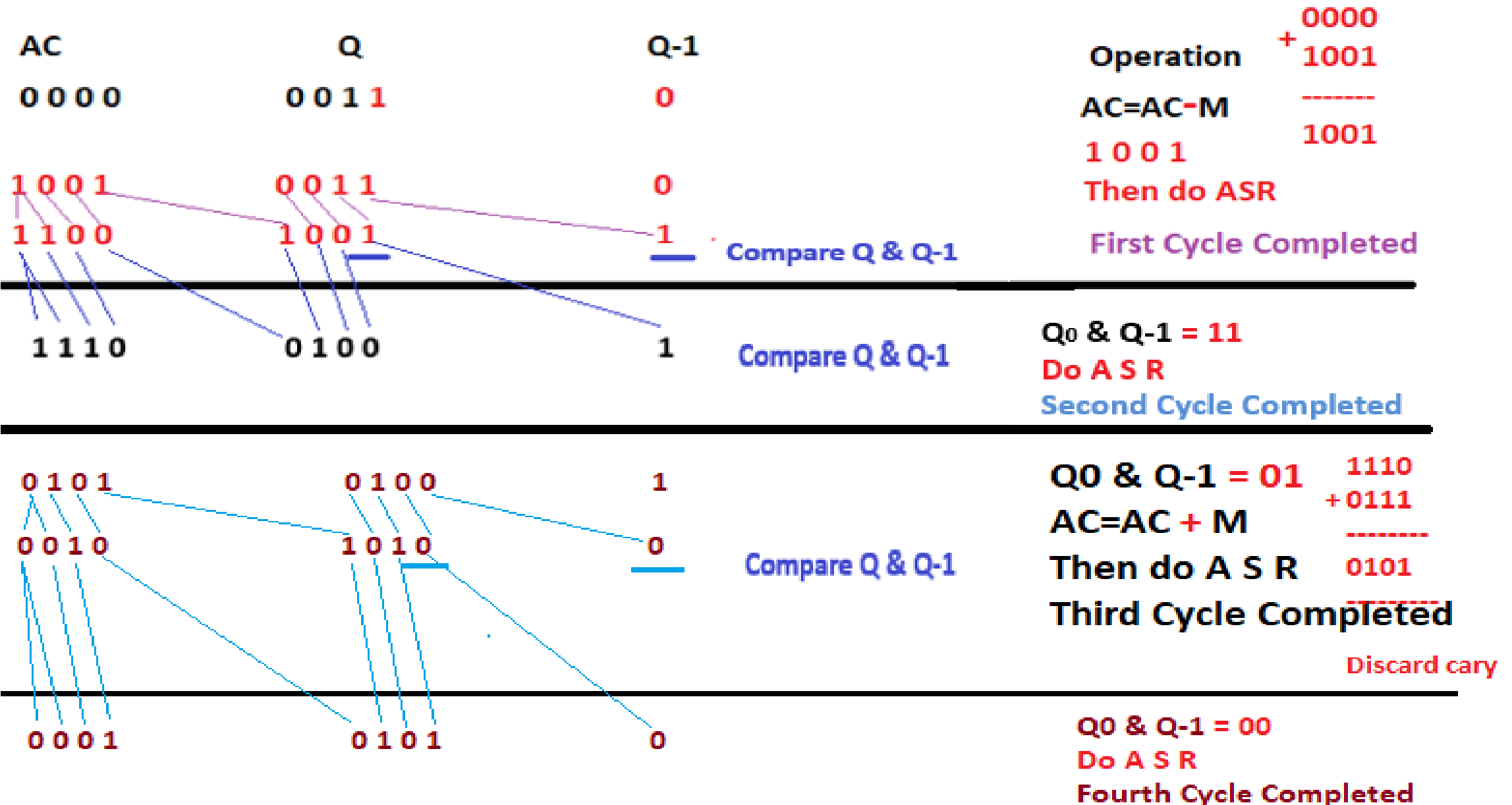
To find - M Binary Value we need to take 2's compliment.

Now Take Multiplicand M value 0 1 1 1

Find 1's compliment 1 0 0 0

Find 2's compliment 1

This is -M value -----
 1 0 0 1



21 in binary

2	21	 1
2	10	 0
2	5	 1
2	2	 0
	1	

$$\therefore 21_{10} = \underline{10101}_2$$